

Air Canada 143

Flight model: Boeing 767
Date: July 23, 1983

From: Montreal , Canada
To: Edmonton, Canada

Boeing 767 is one among those which at that time had most of the controls managed by the microprocessor such that the job of the flight engineer was not needed. The crew was new to the flight model and the flight system. Boeing 767 is a 2 engine plane.

Problem

In the mid journey, the crew receives a warning of low pressure in one of the fuel pump. The 767 has 3 fuel tanks, 2 on the wings and one in the center which is used in long distance flights. Low pressure means one of the pumps need maintenance. Then another low fuel pressure warning in the other side of fuel tank. The digital gauges showing the fuel levels are out of service. 2 pump failure in a brand new plane was unusual. If the left tank is low then, the right tank may be low as well. They decide to land the plane before they run out of fuel and the nearest airport is Winnipeg. Soon after they lose their left engine, which is a clear indication of losing fuel. They soon find a probability of losing another engine. Soon the other engine also shuts down. All flight instruments become blank. They were only trained to manage a single engine failure. They usually become uncontrollable. But, in modern plane to handle such situation a small propeller is deployed called **RAT** in the bottom of the plane body. This is primarily used to generate power for hydraulic system and other basic system. They declare mayday. The ATC loses the transponder signal that is used to track the position of the plane for them to guide which is a part of the secondary radar. This is because the RAT does not power the transponder. The pilots do not have information on the speed and rate of descent to calculate the glide. Then the ATC turns from **True radar (Modern radar)** to **Reflective radar (Old fashion radar)** that does not want the transponder signal. Using this the ATC is able to provide rough calculations of the distance from the airport. The pilots discover that Gimili, the nearest airbase which is much closer and one of the pilots being trained in the airbase for armed forces they knew the base well. But Gimili has no emergency equipment and no ATC there. One of the pilots calculates that they might fall 15 miles before the runway. As there is no power to power the hydraulics of the landing gear, the landing gear drops down due to its own weight this is called as **Gravity drop** of the landing gear. The two main wheels are heavy and get locked properly. The nose gear which is light weighted does not get locked properly. The plane is too high for it to land. If the plane follows a normal approach it would be short of the runway. If it maintains this altitude, when landing it might be over speeding which would overshoot the runway. They try to perform the maneuver called **Slide slip** which is common among gliders and usually not used for planes. Slide slip involves dipping the plane sideways without increasing the airspeed. Unfortunately, The Gimili runway is no longer used to land plane and used by sport car club. The plane manages to land safely on ground with the main gear and the front of the plane scraping the road.

Investigation

There was no evidence of any fuel leaks. When the fuel levels in the tanks were checked only minimal quantity of fuel is left. This brings to the conclusion the flight took off with fuel less than required for its journey. Upon looking at the computer systems in the plane it was found that the **Digital**

fuel gauge processor (Honeywell's processor) had failed. There was no spare in Montreal so it couldn't be replaced. It was found that the plane took off with less than half the fuel that was necessary was filled at the time of takeoff. The fuel refilling unit was aware of the problem and had preformed a manual check of the fuel level. But, the problem is that the Fuel required is calculated by weight of fuel by the pilot and the fuel refilling unit calculates by volume. This conversion is done by the fuel refilling unit. But, this conversion is done in pounds instead of kilograms. In Canada, the fuel was converted into pound but the metric gauges in the 787 requires the fuel in kilograms. So instead of 22,300 kg, the fuel was calculated as 22,300 lbs (pound) which is equivalent of 11,130 kgs of fuel. In the past fuel was calculated by the flight engineer. But, here the flight engineer was been replaced.

Changes that were brought after the incident

The investigation concluded that the pilots and the ground staff were not given enough training in metric calculation involved. The investigation board directed Air Canada to give sufficient training for its staff on the same. And also recommend the airline to keep more spare parts such as fuel gauge processors.

Reference

1. Air Canada 143 Becomes Powerless And Falls From The Sky — Boeing 767 — Mayday: Air Disaster
<https://www.youtube.com/watch?v=9wnja3h70DM>